Title: Network Connectivity Troubleshooting and Command-Line Diagnostics

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1. Executive Summary & Objective

One of the most frequent tickets a Helpdesk Technician receives is a user stating they have lost connection to the network or cannot reach internal server. This document details how to diagnose a broken network adapter from the Windows command prompt, trace the root cause of a connection drop, and restore proper communications using standard network utilities.

2. Environmental Specifications

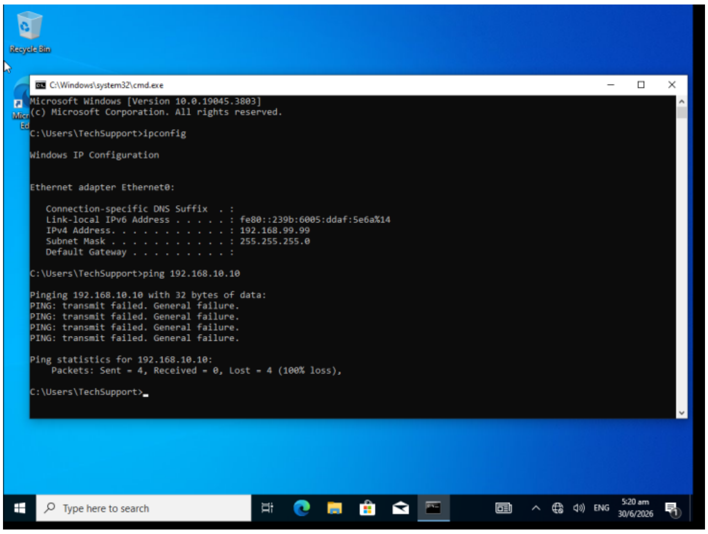
- **Core Server Node**: Windows Server 2022 Domain Controller (192.168.10.10)
- **Target Endpoint**: Windows 10 Pro Workstation
- **Diagnostic Terminal**: Windows Command Prompt (cmd)

3. Creating the Network Failure (Simulating the Issue)

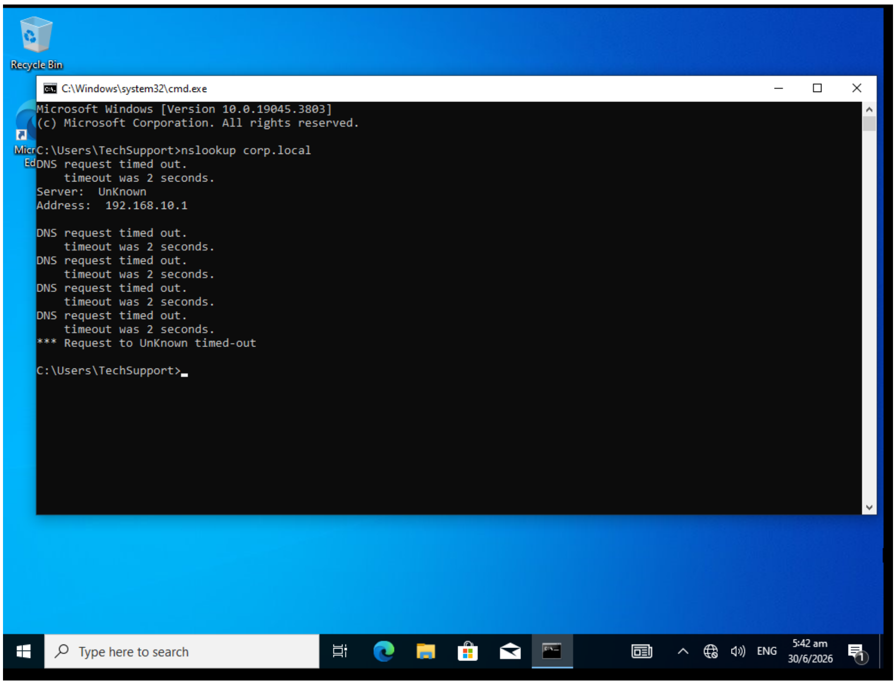
To practice real-world diagnostics, the network settings on the Windows 10 workstation were manually broken to simulate an accidental static IP misconfiguration or configuration mismatch.

4. Confirming the Connection drop via Terminal

1. Attempt to test data packet delivery to the central server by running “ping” command.



2. Because the client workstation is on the wrong subnet, data packets cannot reach the server, resulting in a “**Transmit failed. General failure**”.
3. To track down exactly why the computer cannot communicate, run a **DNS lookup command** – “nslookup corp.local”. Test if the computer can translate server name into IP address.

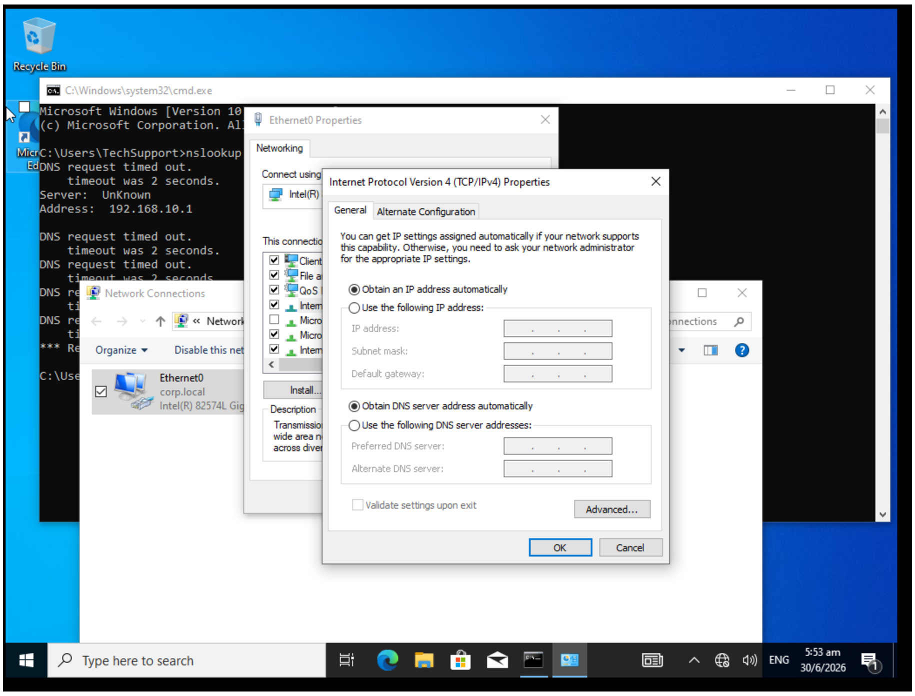


4. This failure confirms a dual problem, **invalid IP address**, and **incorrect network DNS** setting.

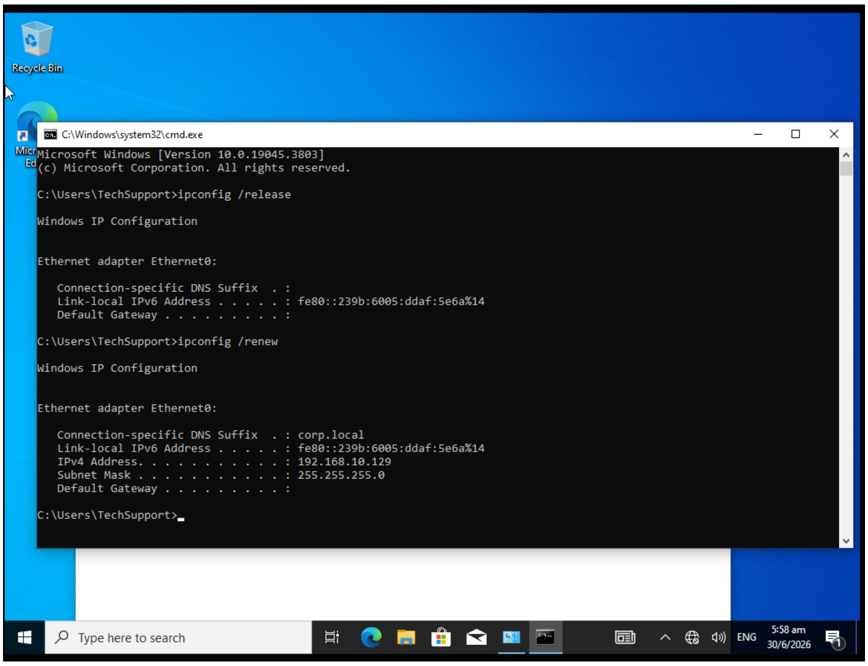
4. Repairing the Connection and Restoring Access

To resolve the issue, the client computer must return to automatic deployment so it can request a clean configuration from the server's active DHCP pool.

1. Open the run box (Win + R), type "ncpa.cpl" to bring up the **Network Connection** window.
2. Right-click on the **Ethernet**, and select **Properties**.
3. A Ethernet Properties window will pop-up.
4. Double-click Internet Protocol Version (TCP/IPv4), a IPv4 window will pop-up.
5. Change both the IP and DNS settings to "Obtain an IP address automatically" and "Obtain DNS server address automatically" and click OK.



6. Return back to the command prompt terminal and force the client machine to drop its old broken identity and request a fresh, verified network identity from the server.
7. Type “ipconfig /release” follow by “ipconfig /renew”.



8. Verify the new setting by running ipconfig, it should show the new identity IP address.
9. Run final confirmation test: “ping 192.168.10.10” and “nslookup corp.local”.
10. The server now should response with active replies, translate corp.local to ip address, proving the connection is fully repaired.